# CHRISTOS LIOSIS

Physicist · PhD in Computational Fluid Dynamics · Postdoctoral Researcher & Academic Fellow, University of West Attica

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## PROFILE

Physicist working on computational fluid dynamics, with a focus on magnetic nanoparticles, micromixers and Tesla valves for water purification and biomedical applications. 3D simulations in OpenFOAM combined with experiments, additively manufactured sensors and machine learning. Member of the Flow Analysis and Simulation Team (UNIWA) since 2015.

## EDUCATION

**2023** **PhD in Computational Fluid Dynamics**
*University of Thessaly, Department of Civil Engineering*
Thesis: *Modeling for Water Purification from Heavy Metals Using Microfluidic Devices*.
Supervisor: Prof. T. Karakasidis

**2018** **MSc in Energy Technology**
*Heriot-Watt University – University of West Attica*
Thesis: *Numerical study of nanoparticles mixing in micropipes*. Supervisor: Prof. I. Sarris

**2017** **MSc in STEM Education**
*School of Pedagogical and Technological Education (ASPETE)*
Thesis: *Developing S.T.E.M. experiments using Arduino, Matlab and their didactic analysis*

**2015** **Postgraduate Diploma in Materials Science for Advanced Technologies**
*Universities of Ioannina, Patras and Crete*

**2014** **BSc in Physics**
*University of Ioannina*

## RESEARCH EXPERIENCE

**2024–now** **Postdoctoral Researcher**
*Department of Mechanical Engineering, University of West Attica*
Models for guiding, mixing and separating magnetic nanoparticles for water purification and biological applications.

**2019–2023** **Doctoral Researcher**
*Department of Civil Engineering, University of Thessaly*
Heavy-metal removal with magnetic nanoparticles in passive and magnetically driven mixers (OpenFOAM, ParaView).

**2015–now** **Member, Flow Analysis and Simulation Team**
*University of West Attica (fluids.uniwa.gr)*

**2014–2018** **Master Researcher**
*Department of Energy Technology, University of West Attica*
Mixing of magnetic nanoparticles under external magnetic fields (OpenFOAM).

## TEACHING

**2023–2026** **Electromagnetism and AC Circuit Analysis (NMB.206)**
*Department of Biomedical Engineering, University of West Attica · spring semesters*

**2024–2025** **Introduction to Fluid Dynamics (MM109E03)**
*Department of Mechanical Engineering, University of West Attica*

## RESEARCH PROJECTS

**2026–2028** **SEEU – Small enhanced European UAS**
*European Defence Fund, EDF-2024-DA-C4ISR-SEEU-STEP*

**2025–2028** **Verdedrive**
*HORIZON-MSCA Staff Exchanges 2023*
By-products for eco-friendly and sustainable diesel fuel.

**2024–2027** **PROMATAI**
*HORIZON-MSCA Staff Exchanges 2022*
Processing of hybrid materials and nanomaterials using AI algorithms.

**2024–2027** **ST3AM**
*Erasmus+ KA220-HED*
Gender balance, education reformation and mentorship in STEM.

**2023–2026** **AGRHI**
*Erasmus+ CBHE, ERASMUS-EDU-2023-CBHE-STRAND-2*
Agricultural Revitalization in Higher Education Institutions of Asian Countries.

**2020–2021** **Magnetic navigation of nanoparticles in real networks of human arteries**
*MIS 5048923*

## AWARDS & SCHOLARSHIPS

**2025–2026** **ARIS allocation (GRNET) – Computational Physics Networks**
150,000 CPU core hours and 50,000 GPU hours

**2024–2025** **ARIS allocation (GRNET) – Homogeneous Turbulence Model**
100,000 CPU core hours

**2023** **Young Researcher Award**
*ICRAFMN-2023, Bengaluru, India*

**2021** **Best Paper Award**
*Int. Conf. on Drug Delivery Nanosystems for Biomedical Engineering Applications, Rome*

## JOURNAL ARTICLES

1. D.N. Pagonis, **C. Liosis**, A. Vailas, D. Zagklaras, S. Dimitrellou and E. Strantzali, «Additively Manufactured Ring-Type Thermal Sensor for In-Pipe Flow Monitoring in a Marine Engineering Context: Design Evolution and Electrothermal Characterisation», *Sensors*, 2026, 26(14), 4586. doi:10.3390/s26144586
2. **C. Liosis**, D.N. Pagonis, S. Peppa, M. Drossos and I. Sarris, «3D-Printed Tesla Valve with IoT-Based Flow and Pressure Sensing», *Fluids*, 2026, 11(3), 69. doi:10.3390/fluids11030069
3. **C. Liosis**, A. Papadatos, D.N. Pagonis, S. Peppa and I. Sarris, «Diodicity of MicroTesla Valves Under Various Re Numbers», *Micromachines*, 2025, 16, 1329. doi:10.3390/mi16121329
4. **C. Liosis**, G. Sofiadis, E. Karvelas, T. Karakasidis and I.E. Sarris, «Inverse Tesla Valve as Micromixer for Water Purification», *Micromachines*, 2024, 15, 1371. doi:10.3390/mi15111371
5. **C. Liosis**, G. Sofiadis, E. Karvelas, T. Karakasidis and I.E. Sarris, «A Tesla Valve as a Micromixer for $Fe_3O_4$ Nanoparticles», *Processes*, 2022, 10(8), 1648. doi:10.3390/pr10081648
6. **C. Liosis**, E. Karvelas, T.E. Karakasidis and I.E. Sarris, «Mixing of $Fe_3O_4$ Nanoparticles under Electromagnetic and Shear Conditions for Wastewater Treatment Applications», *Journal of Water Supply: Research and Technology – AQUA*, 2022, 71(6), 671–681. doi:10.2166/aqua.2022.080
7. **C. Liosis**, A. Papadopoulou, E. Karvelas, T.E. Karakasidis and I.E. Sarris, «Heavy Metals Adsorption Using Magnetic Nanoparticles for Water Purification: A Critical Review», *Materials*, 2021, 14(24), 7500. doi:10.3390/ma14247500

8. E. Karvelas, **C. Liosis**, A. Theodorakakos, I.E. Sarris and T.E. Karakasidis, «An Optimized Method for 3D Magnetic Navigation of Nanoparticles inside Human Arteries», *Fluids*, 2021, 6(3), 97. doi:10.3390/fluids6030097
9. **C. Liosis**, E.G. Karvelas, T.E. Karakasidis and I.E. Sarris, «Numerical Study of Magnetic Particles Mixing in Water under an External Magnetic Field», *Journal of Water Supply: Research and Technology – AQUA*, 2020. doi:10.2166/aqua.2020.090
10. E.G. Karvelas, **C. Liosis**, L. Benos, T.E. Karakasidis and I.E. Sarris, «Micromixing Efficiency of Particles in Heavy Metal Removal Processes under Various Inlet Conditions», *Water*, 2019, 11, 1135. doi:10.3390/w11061135

## BOOK CHAPTERS

1. N.D. Polychronopoulos, K. Moustris, T. Karakasidis, E. Karvelas, **C. Liosis**, G. Sofiadis, P. Pimenidou, S. Peppa and I. Sarris, «Random Forest Machine Learning Algorithm for Screw Design in Polymer Extrusion», in *Development and Testing of Innovative Solutions for the Processing of Hybrid Materials and Nanomaterials Using Artificial Intelligence Algorithms: Monograph, Volume I*, edited by L. Dulebova and V. Krasinskyi, Technical University of Kosice, Kosice, 2024, pp. 33–46. ISBN 978-80-553-4741-7.

## PATENTS

1. D.N. Pagonis, **C. Liosis** and N. Sgouros, «Additively Manufactured Multifunctional Electrical Interconnect and Sensing Structure», Hellenic Industrial Property Organisation (OBI), patent application filed June 2026 – pending.

## CONFERENCE PROCEEDINGS

1. G. Sofiadis, **C. Liosis**, T. Karakasidis and I. Sarris, «Reynolds Number Effect on the Turbulent Micropolar Open-Channel Flow with Sediment Transport», *Environmental and Earth Sciences Proceedings*, 2026, 44(1), 23. https://www.mdpi.com/3042-5743/44/1/23
2. J. Lawen, V. Gallelli, **C. Liosis**, I. Sarris and S. Peppa, «Automatic Voronoi-Based Meshing for CFD Analysis of Passive Flow Control in Diodic Pipes», *MATEC Web of Conferences*, 2026, 421, 03002. doi:10.1051/matecconf/202642103002
3. **C. Liosis**, G. Sofiadis, E. Karvelas, T. Karakasidis and I. Sarris, «Simulations of Tesla Valve Micromixer for Water Purification with $Fe_3O_4$ Nanoparticles», *Environmental Sciences Proceedings*, 2022, 21(1), 82. doi:10.3390/environsciproc2022021082
4. E.G. Karvelas, **C. Liosis**, T.E. Karakasidis and I.E. Sarris, «Mixing Nanoparticles and Contaminated Water Under Different Velocities for Optimum Heavy Metal Ions Adsorption», *Environmental Sciences Proceedings*, 2020, 2, 65. doi:10.3390/environsciproc2020002065
5. E.G. Karvelas, **C. Liosis**, T.E. Karakasidis and I.E. Sarris, «Mixing of Nanoparticles in Micromixers Under Different Angles and Velocities of the Incoming Water», *Proceedings*, 2018, 2, 577. doi:10.3390/proceedings2110577

## CONFERENCE PRESENTATIONS & POSTERS

1. G. Sofiadis, V. Perivolioti, **C. Liosis**, I. Sarris, C. Manopoulos, A. Raptis and E. Karvelas, «Uncertainty quantification of steady-state flow in a three-dimensional arterial model (in Greek)», $13^{th}$ Panhellenic Conference on Fluid Flow Phenomena (ROI 2026), Volos, Greece, 4–5 December 2026.
2. **C. Liosis**, D.N. Pagonis, T. Karakasidis and I. Sarris, «ATtiny-Based Laser Control System for Engraving 3D-Printed Sensor Structures», Micro Nano & Chips Tech 2026, $13^{th}$ International Conference on Micro-Nanoelectronics, Nanotechnology and MEMS, Athens, Greece, 15–18 October 2026.
3. K. Moustris, **C. Liosis**, I. Sarris, G. Spyropoulos, N. Polychronopoulos, G. Sofiadis and E. Karvelas, «Machine Learning-Based Prediction of Feed Section Length in Single-Screw Extruders Using Rarity-Weighted Data Augmentation on a Small Industrial Dataset», $1^{st}$ MAHSA International Conference on Sustainable Engineering, Built Environment, and Information Technology (MI-SEBIT 2026), Malaysia, 25–26 September 2026.
4. J. Lawen, V. Gallelli, **C. Liosis**, I. Sarris and S. Peppa, «Automatic Voronoi-Based Meshing for CFD Analysis of Passive Flow Control in Diodic Pipes», $1^{st}$ International Conference on Monitoring and Control of Water Systems (MOCWS 2026), Chalkida, Greece, 28–29 May 2026.

5. G. Sofiadis, **C. Liosis**, T. Karakasidis and I. Sarris, «Reynolds Number Effect of the Turbulent Micropolar Open Channel Flow with Sediment Transport», 6th EWaS International Conference, Thessaloniki, Greece, 11–14 May 2026.
6. V. Krasinskyi, K. Bajer, A. Bartnicki, L. Dulebova, E. Karvelas and **C. Liosis**, «Projektowanie i optymalizacja uniwersalnej strefy zasypu jednoślimakowej wytłaczarki (in Polish)», 22nd Engineering Polymers and Composites, Silesian University of Technology, Szczyrk, Poland, 21–24 October 2025.
7. K. Moustris, N. Polychronopoulos, P. Pimenidou, G. Spyropoulos, I. Sarris, G. Sofiadis, **C. Liosis**, E. Karvelas and S. Peppa, «Application of Artificial Intelligence and Machine Learning Classification Models for Predicting Screw Geometry Parameters in Single-Screw Extruder», 14th International Conference on Mathematical Modeling in Physical Sciences, 20–23 October 2025.
8. D.N. Pagonis, D. Zagklaras, A. Vailas, I. Iakovidis, S. Dimitrellou and **C. Liosis**, «Additively Manufactured Thermal Sensor for Local Temperature Monitoring: Experimental and Numerical Validation», 22nd International Conference on Manufacturing Research, University of Birmingham, UK, 10–12 September 2025.
9. T. Karakasidis, E. Karvelas, G. Sofiadis, **C. Liosis** and I. Sarris, «Nanoparticles and Blood Mixing in Microfluidic Reactors using Computational Fluid Dynamics», 21st International Conference on Nanosciences & Nanotechnologies (NN24), Thessaloniki, Greece, 2–5 July 2024.
10. **C. Liosis**, G. Sofiadis, E. Karvelas, T. Karakasidis and I.E. Sarris, «Tesla Valve as Micromixer for Water Purification with Magnetic $Fe_3O_4$ Nanoparticles», International Conference on Recent Advances in Fluid Mechanics and Nanoelectronics (ICRAFMN-2023), Bengaluru, India, 12–14 July 2023 – **Young Researcher Award**.
11. **C. Liosis**, G. Sofiadis, E. Karvelas, T. Karakasidis and I.E. Sarris, «Simulations of Tesla Valve Micromixer for Water Purification with $Fe_3O_4$ Nanoparticles», 5th EWaS International Conference, Naples, Italy, 12–15 July 2022.
12. E. Karvelas, **C. Liosis**, T. Karakasidis and I.E. Sarris, «Magnetic Navigation of Nanoparticles for Drug Delivery inside Carotid Artery Under the Cardiac Cycle», XXXV Panhellenic Conference on Solid State Physics and Materials Science, NCSR «Demokritos», Athens, Greece, 26–29 September 2021.
13. E.G. Karvelas, **C. Liosis**, A. Theodorakakos and T.E. Karakasidis, «Magnetic Navigation of Nanoparticles Inside a 3D Carotid Model», International Conference on Drug Delivery Nanosystems for Biomedical Engineering Applications, Rome, Italy, 18–19 January 2021 – **Best Paper Award**.
14. E.G. Karvelas, **C. Liosis**, T.E. Karakasidis and I.E. Sarris, «Mixing Nanoparticles and Contaminated Water Under Different Velocities for Optimum Heavy Metal Ions Adsorption», 4th EWaS International Conference, Greece, 24–27 June 2020.
15. E.G. Karvelas, **C. Liosis**, T.E. Karakasidis and I.E. Sarris, «Particle mixing using magnetic fields in micropipes (in Greek)», 12th Panhellenic Scientific Conference of Chemical Engineering, NTUA, Athens, Greece, 29–31 May 2019.
16. E.G. Karvelas, **C. Liosis**, T.E. Karakasidis and I.E. Sarris, «Mixing of Nanoparticles in Micromixers Under Different Angles and Velocities of the Incoming Water», 3rd EWaS International Conference, Lefkada, Greece, 27–30 June 2018.
17. **C. Liosis**, A. Papadopoulou and E. Liosi, «Simple pendulum and measurement of the acceleration of gravity using video analysis (in Greek)», 5th Neos Paidagogos Conference, Eugenides Foundation, Athens, Greece, 28–29 April 2018.
18. **C. Liosis**, «Use of Arduino in vehicle braking (in Greek)», 5th International and Interscientific Conference eLife 2016, Plastiras Lake, Greece, 23–26 June 2016.

## ACADEMIC SERVICE

- **Guest Editor**, *Symmetry* (MDPI), special issue «Advances of Symmetry/Asymmetry in Fluid Dynamics: Engineering Applications and Marine Hydrodynamics» (2025)
- **Language Editor**, Kallipos Open Academic Editions (2024): *Computational Fluid Dynamics* (Sarris, Karvelas & Sofiadis); *Embedded Systems Implementation for Networks and Applications in Linux* (Milionis); *Computational Chemistry and Molecular Modeling* (Sigalas, Koukaras & Charistos)
- **Co-supervisor**, Erasmus+ traineeships UNIWA – CESI (France): J. Kolo (heat exchangers for heat pumps), A. Ekelle (energy balance of non-interconnected islands)

- **Undergraduate thesis committees**, University of West Attica (wind turbine CFD; Tesla valve performance; ML for aerodynamic coefficients)

## CONTINUING EDUCATION

**06/2023** **BioMed AI Summer School**
*Academy of Athens*

**2020–2021** **Expert in Finite Element Analysis**
*National and Kapodistrian University of Athens*

**2017** **LaTeX; Programming with Python**
*University of Piraeus; Aristotle University of Thessaloniki*

**2016** **Arduino**
*Hellenic Physical Society*

**11/2015** **National Training Programme of Greek Physicists**
*CERN*

## SKILLS

OpenFOAM, ParaView, gmsh, Finite Element Analysis, Python, Matlab, Mathematica, Machine Learning, LaTeX, Arduino, 3D printing, CAD

## LANGUAGES & MEMBERSHIPS

Greek (native), English (C2), Spanish (B1).
Hellenic Physical Society · European Physical Society · Hellenic Education Society of STEM